

Uday Kiran Battula

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Education

Amrita Vishwa Vidyapeetham – B.Tech in Computer Science – 7.57/10.0

Aug 2021–May 2025

Experience

Mindcres Technologies | AI/ML Engineer

Mar 2026–Present

- Architecting voice + web grievance bot for AP Government's PGRS portal — a 25-state conversation state machine handling multilingual (Telugu/English) flows over Google Gemini, Sarvam AI, and Exotel WebSocket telephony.
- Building modular FastAPI backend (async, asyncpg) split into 4 microservices — admin, bot, IVR, verification — with provider-swap toolkit abstractions for LLM, OCR, Speech, and Audio enabling zero-change vendor swaps.
- Designing YAML-driven prompt management, scoped API-key + B2B widget-key auth with Redis rate limiting, and federated admin log viewer; refactoring a 7,000+ line monolith while maintaining 2,300+ Pytest cases at 90%+ coverage on Dokku, PostgreSQL, and Redis with Langfuse observability.

Spinnaker Analytics (US-Based, Remote) | AI Engineer Intern

Sep 2025–Feb 2026

- Architected 8-stage RAG pipeline over 69 healthcare documents (345 chunks) using sentence-transformers + FAISS, hitting 93.3% retrieval precision and 76% validation rate with section re-ranking, token-overlap validation, and a 25% confidence threshold for safety-first clinical decision support.
- Designed Equity Filings Agent with adapter-pattern loaders and lexicon-based scoring to automate 10-K/10-Q analysis — structured reports in seconds vs hours of manual reading, with strict grounding prompts cutting false-positive validations by prioritizing precision over recall.
- Built a Model Context Protocol (MCP) server using FastMCP over STDIO transport exposing 3 agentic tools with SQLite persistence and Claude Desktop integration, enabling protocol-level LLM-tool invocation from natural-language intent without hardcoded validation.

Projects

Real-Time E-Commerce Voice Bot

- Developed full-stack voice assistant in React + Web Speech API for 4 intents (product search, order tracking, returns, recommendations) with a classification router and explicit tool execution preventing hallucinated order IDs and product names.
- Architected ChromaDB RAG with metadata filtering on price and category over a 500-item, 30+ category inventory; LangSmith observability tracing all LLM calls, retrievals, and tool invocations for audit and quality monitoring.

Car Damage Detection & Mobile Deployment

- Trained a 6-class vehicle-damage classifier (cracks, dents, scratches, glass shatter, flat tyres, lamp breakage) on a Kaggle dataset of ~400 images per class, fine-tuning a pretrained DenseNet-169 backbone with a custom classifier head — raising baseline CNN accuracy from 82% to 95% via data augmentation, hyperparameter tuning, and per-class confusion-matrix analysis.
- Exported the trained model to TensorFlow Lite and shipped it inside a native Android app (Kotlin, Android Studio) for on-device offline inference, enabling field damage assessment for insurance-claim automation without server round-trips.

Certifications

Introduction to Generative AI — Google Cloud Skills Boost

Generative AI and Agentic AI Development — Boston Institute of Analytics

Machine Learning with Python — Infosys Springboard

Technical Skills

Programming Languages: Python, SQL, JavaScript, Java, Kotlin

Generative AI & Large Language Models: Google Gemini, OpenAI, Groq & Claude APIs, LangChain, LiteLLM, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI Systems, Model Context Protocol (MCP), Claude Code

Machine Learning, Deep Learning & Vision: TensorFlow, Keras, PyTorch, Scikit-learn, CNN, Transfer Learning, DenseNet, TensorFlow Lite, NLP

Retrieval & Speech: FAISS, ChromaDB, BM25, sentence-transformers, Gemini Embeddings, Sarvam AI (Telugu ASR/TTS), Exotel WebSocket

Backend & Databases: FastAPI, async/asyncpg, SQLAlchemy, Alembic, Pydantic, FastMCP, PostgreSQL, Redis

DevOps, Observability & Testing: Docker, Dokku, Git, GitHub Actions, Langfuse, LangSmith, Pytest, Android Studio